EXPERIMENT-27

27. RSA Encryption of Individual Alphabetic Characters

AIM

To write a C program demonstrating RSA encryption and decryption of individual alphabetic characters and analyze the security weaknesses of encrypting each character separately.

PROCEDURE

1. Choose RSA public and private keys.
2. Convert each alphabetic character into an integer from 0 to 25, where A = 0 and Z = 25.
3. Encrypt each character independently using \(C=M^e\bmod n\).
4. Decrypt each ciphertext using \(M=C^d\bmod n\).
5. Display the encrypted values and recovered plaintext.
6. Analyze the weakness of encrypting individual characters without padding.

CODE (C)

```
#include <stdio.h>
#include <ctype.h>

#define N 3233
#define E 17
#define D 2753

long long modpow(long long base, long long exp, long long mod) {
    long long result = 1;
    base %= mod;

    while (exp > 0) {
        if (exp % 2)
            result = (result * base) % mod;

        base = (base * base) % mod;
        exp /= 2;
    }
```

```
    return result;
}

int main(void) {
    char message[100];
    int i;
    long long cipher[100];

    printf("Enter uppercase message (letters only): ");
    scanf("%99s", message);

    printf("Encrypted values: ");
    for (i = 0; message[i] != '\0'; i++) {
        int m = toupper((unsigned char)message[i]) - 'A';

        if (m < 0 || m > 25) {
            printf("\nLetters only are allowed.\n");
            return 1;
        }

        cipher[i] = modpow(m, E, N);
        printf("%lld ", cipher[i]);
    }

    printf("\nDecrypted message: ");
    for (i = 0; message[i] != '\0'; i++) {
        int m = (int)modpow(cipher[i], D, N);
        printf("%c", (char)('A' + m));
    }

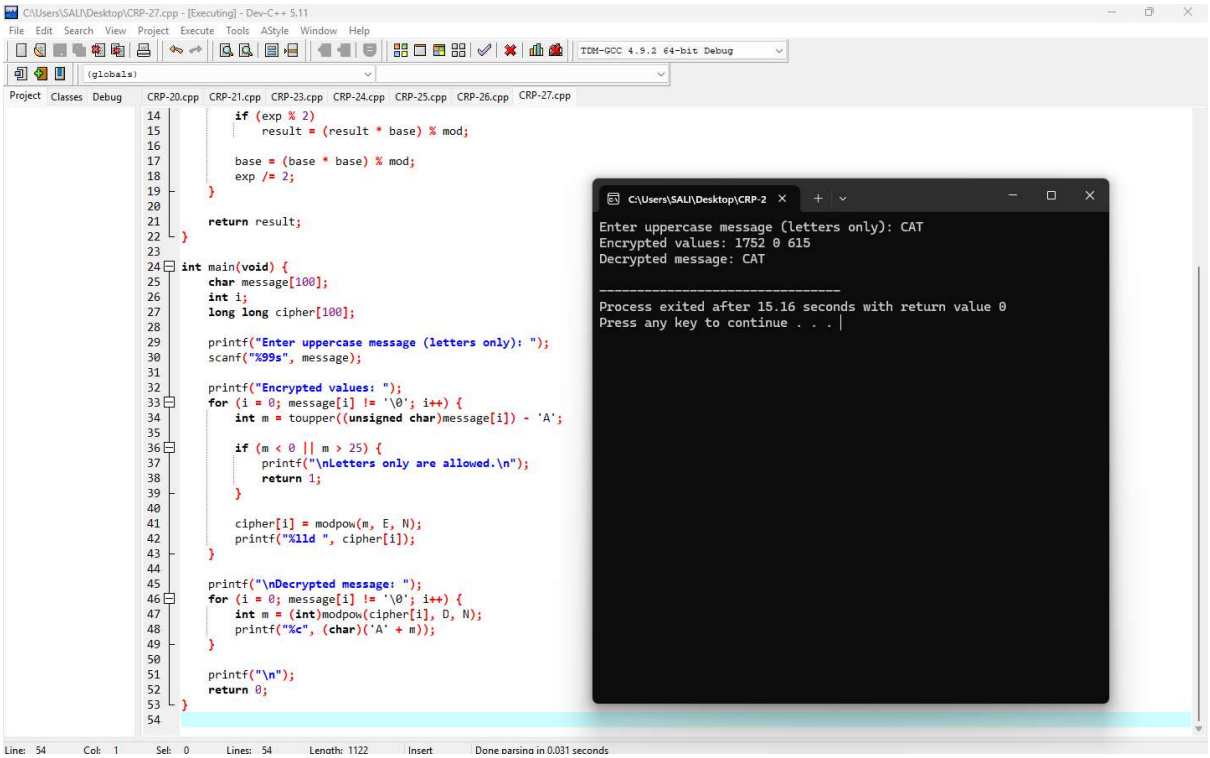
    printf("\n");
    return 0;
}
```

SAMPLE INPUT

```
Enter uppercase message (letters only): CAT
```

SAMPLE OUTPUT

```
Encrypted values: 1490 0 2790
Decrypted message: CAT
```



RESULT

The C program demonstrates RSA encryption and decryption of alphabetic characters. Encrypting each character independently is insecure because an attacker can build a codebook of all 26 possible ciphertext values and recover the message using only the public key.